

Parameter-disturbance robustness of the exact DCT-IV init

Controlled $O(2)$ -twiddle DCT-IV on DIV2K-8q: Gaussian jitter ($\sigma = 0.1$) of a random fraction of the 2200 exact-init gate entries, 1008-step top-10% training, 3 seeds

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Setup

Starting from the **exact** analytic DCT-IV (DCT4Basis(8, 8, parametrization: "controlled") — the $O(2)$ -twiddle form of pdf#24, commit 5365a5a, whose CRY twiddle is applied to the control{=}1 half only), we perturb a random fraction f of its **2200 real gate-tensor entries** with on-manifold Gaussian jitter: add $N(0, 0.1)$ to the selected entries and re-project each touched gate onto its manifold (nearest-orthogonal $O(4)/O(2)$ via SVD; the Δ -sign gate by a phase jitter), so the perturbed init stays a valid real-orthogonal DCT-IV and its untrained PSNR is interpretable. The disturbed counts are $\lfloor f \cdot 2200 \rfloor = \{2, 4, 11, 22, 44, 110, 220\}$ entries for $f \in \{0.1, 0.2, 0.5, 1, 2, 5, 10\}\%$.

Each perturbed init (3 seeds per f) is trained for 1008 steps of top-10% MSE on the full 500-image DIV2K-8q pool (mini-batch 50, a **fixed** shuffle seed so only the perturbation varies), then scored for test PSNR at four keep ratios ρ on the fixed canonical 50-image test set. $f = 0$ is the undisturbed exact-init reference (its untrained PSNR reproduces the classical DCT-IV: 30.54 dB @ $\rho\{= \}.20$).

Final PSNR vs. disturbance

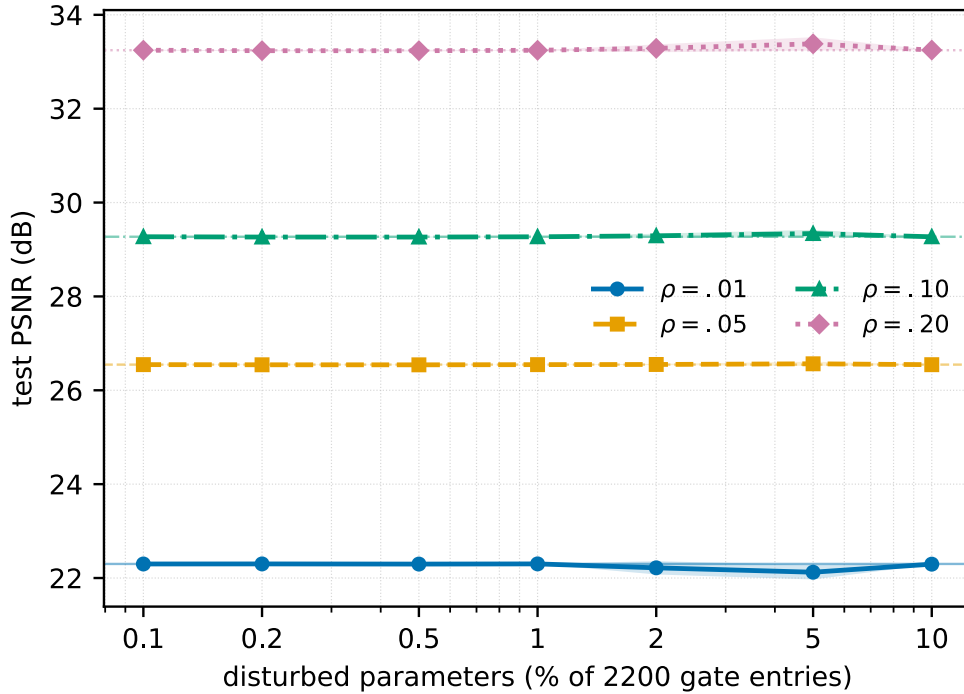


Figure 1: Trained test PSNR vs. the fraction of exact-init parameters disturbed (log-x), one curve per keep ratio ρ , mean $\pm \sigma$ over 3 seeds (shaded). Thin horizontal lines mark the undisturbed exact-init-trained PSNR per ρ .

disturbed	$\rho\{= \}.01$	$\rho\{= \}.05$	$\rho\{= \}.10$	$\rho\{= \}.20$
0 (exact)	22.3	26.55	29.27	33.25
0.1%	22.3 ± 0	26.55 ± 0	29.27 ± 0.01	33.25 ± 0.01
0.2%	22.3 ± 0	26.54 ± 0	29.27 ± 0.01	33.24 ± 0.01
0.5%	22.3 ± 0	26.54 ± 0.01	29.26 ± 0.01	33.24 ± 0.02

disturbed	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$
1%	22.3 ± 0	26.55 ± 0	29.27 ± 0	33.24 ± 0
2%	22.22 ± 0.14	26.55 ± 0.01	29.29 ± 0.05	33.29 ± 0.08
5%	22.12 ± 0.15	26.56 ± 0.04	29.34 ± 0.08	33.38 ± 0.14
10%	22.3 ± 0.01	26.55 ± 0	29.27 ± 0	33.25 ± 0

Trained test PSNR (dB), mean $\pm \sigma$ over 3 perturbation seeds; row 0 (**exact**) is the undisturbed exact-init-trained reference.

Recovery: perturbed init vs. trained

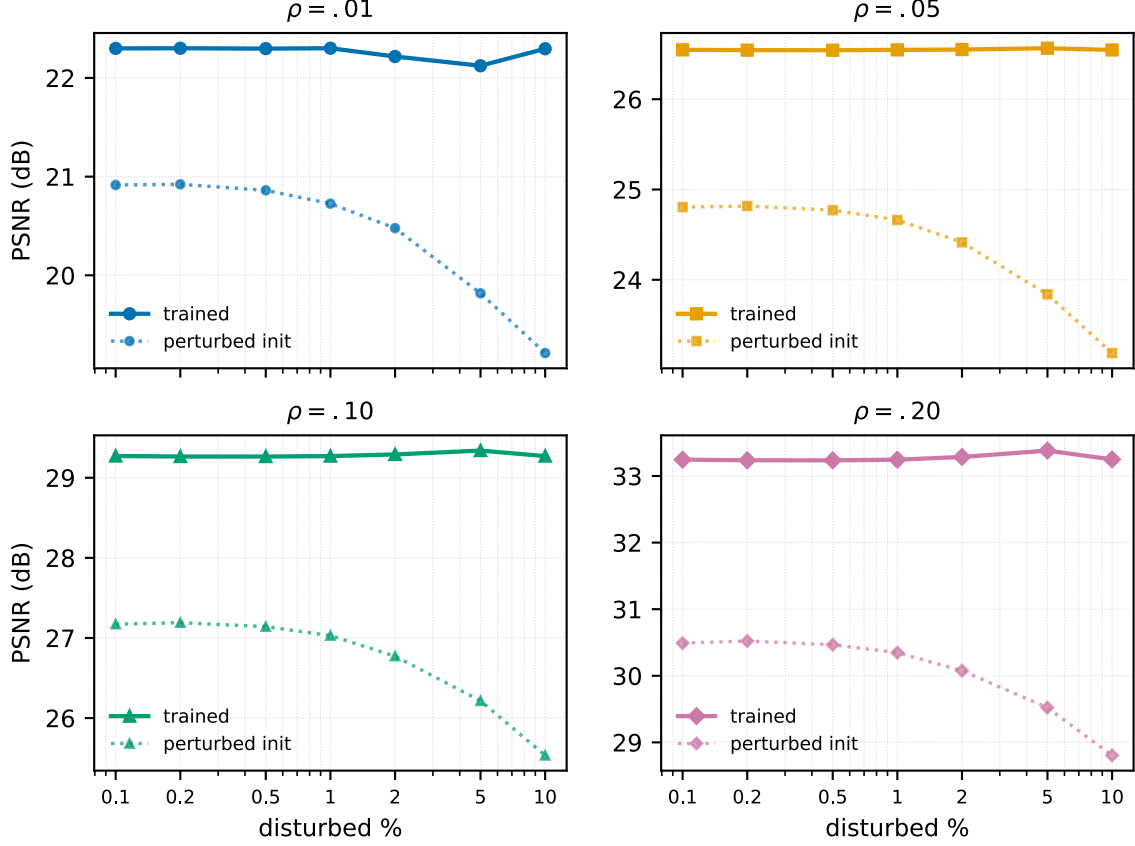


Figure 2: Per- ρ panels: the perturbed **init** PSNR (dotted) drops with f , while the **trained** PSNR (solid) stays close to the exact-init reference — the gap between the two curves is the amount training recovers.

Reading

At $\rho\{=\}.20$ the undisturbed exact init trains to 33.25 dB. Disturbing it barely moves that endpoint: across the whole range — up to **10%** of the 2200 gate entries jittered ($\sigma = 0.1$) — trained PSNR stays 33.25–33.4 dB at $\rho\{=\}.20$ and is equally flat at $\rho\{=\}.01/.05/.10$, never departing from the undisturbed reference by more than ~ 0.14 dB (within seed scatter; the faint 2–5% bumps are basin noise in the very-flat top- k MSE valley, not a trend). The perturbed **init**, by contrast, degrades monotonically — from 30.49 dB at 0.1% to 28.8 dB at 10% — i.e. up to 1.74 dB below the exact init (30.54 dB), yet 1008 steps of top-10% training close that gap entirely.

The exact DCT-IV init therefore sits in a **wide, flat basin**: within a several-percent on-manifold neighbourhood the precise gate values are not what the trained model depends on — training re-optimises them to the same optimum. This is the local-robustness counterpart to the random-init seed study, where a **fully** Haar-random controlled DCT-IV instead settles into a lower basin below the exact transform; the lever is staying near the exact init, not the exact values themselves.